

CUSTOMER SEGMENTATION USING MACHINE LEARNING

An Internship Project Report

Submitted By:

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1. ABSTRACT

This project focuses on customer segmentation using machine learning techniques. The K-Means clustering algorithm is used to group customers based on their annual income and spending score. The segmentation helps businesses understand customer behavior deeply and improve their targeted marketing strategies effectively.

2. INTRODUCTION

Customer segmentation is the process of dividing customers into groups based on similar characteristics. It helps businesses target customers effectively and improve decision making. In this project, machine learning is used to perform segmentation automatically, replacing traditional manual grouping with automated, data-driven analytical insights.

3. OBJECTIVE

The primary objectives of this project are:

- To analyze raw customer data effectively.
- To identify distinct patterns in consumer spending behavior.
- To group customers into clear categories using the K-Means clustering algorithm.

4. DATASET DESCRIPTION

Dataset Used: Mall Customers Dataset

Total Records: 200 rows of customer data

Key Features Included:

- Customer ID
- Gender
- Age
- Annual Income
- Spending Score

5. METHODOLOGY

The implementation of this project follows a structured data science pipeline divided into six core steps:

1. **Data Collection:** Gathering the relevant customer records and traits.
2. **Data Preprocessing:** Cleaning the data fields and formatting them for the model.
3. **Exploratory Data Analysis (EDA):** Finding correlations and viewing data distributions.
4. **Finding Optimal Clusters:** Using the Elbow Method to discover the ideal number of clusters.
5. **Applying K-Means Clustering:** Executing the algorithm on the finalized dataset.
6. **Visualizing Results:** Creating graphical plots to map out and review customer segments.

6. TOOLS USED

The technical ecosystem utilized to develop this project includes:

- **Programming Language:** Python
- **Data Manipulation:** Pandas and NumPy
- **Data Visualization:** Matplotlib and Seaborn
- **Machine Learning:** Scikit-learn

7. ALGORITHM USED

K-Means Clustering

K-Means clustering is an unsupervised learning algorithm used to group similar data points into distinct clusters. It calculates distances between data points and cluster centroids iteratively, optimizing the groupings until the data is organized into non-overlapping sub-groups based on shared properties.

8. RESULTS

By executing the K-Means algorithm, the target customer base was successfully segmented into 5 distinct clusters based on their annual income and spending behavior. Each cluster clearly represents a different customer type or persona, providing actionable business visibility into market groups.

9. FUTURE ENHANCEMENT

The project scope can be expanded in the future through the following initiatives:

- Can be extended using Deep Learning models to handle larger, more complex structural data.

- Can include more customer features such as online browsing footprint and transaction history.
- Can be used for real-time customer prediction by deploying the trained model to active production pipelines.

10. CONCLUSION

Customer segmentation using machine learning helps businesses understand their customers much better and improve marketing strategies effectively. Using data-driven algorithmic models reduces marketing costs and directly enhances customer satisfaction metrics.